Odometry test: step by step

ApexStrykerz · BIOBUZZ 2026–27 · goBILDA Pinpoint with two 4-Bar pods

Our setup: two pods on the long side channels. **X pod**: near a front wheel, its wheel rolls **forward/backward**. **Y pod**: middle of the other side channel, its wheel rolls **sideways**.

You need: gamepad 1, the Driver Station, masking tape, a tape measure, paper and a pen, and a clear space of 4 × 4 ft on the field tiles.

Safety, whenever the robot moves by itself: one person holds **gamepad 1**. **Any button stops the robot.** The red STOP on the Driver Station also works. Nobody stands inside the robot's space; keep hands and feet away from the wheels.

Step 1 · Configure the Pinpoint on the Driver Hub · 5 min

1. Check the cables: the **forward-rolling** pod → Pinpoint **X** port; the **sideways-rolling** pod → **Y** port; the Pinpoint → a Control Hub **I²C port, bus 1** (not bus 0).
2. On the Driver Hub: ⋮ **› Configure Robot**.
3. Select **Apex-1** › **Edit**.
4. Open **Control Hub Portal › Control Hub › I²C Bus 1**.
5. If a device named `pinpoint` is already there, skip to step 8.
6. **Add** a device: type **goBILDA Pinpoint Odometry Computer**, name exactly `pinpoint` (all lower case).
7. **Done › Done › Save** (keep the name **Apex-1**).
8. Make sure **Apex-1** is the **active** configuration. The robot restarts.
9. **Don't touch the robot while it restarts.** The light on the Pinpoint should turn **green**.

Light	Means
Green	Ready
Red	Starting up: keep the robot still
Purple	No pods found: check both pod cables
Blue	X pod not found: check its cable
Orange	Y pod not found: check its cable

Step 2 · Check the pods touch the floor · 2 min

Robot on the floor: each pod's wheel touches the tile with its arm **partly pushed up**. Lift the robot: both wheels drop down. If a wheel does not touch the floor, **tell Asif**: the pod needs to sit lower.

Step 3 · Odometry Check · about 10 min

Driver Station › **TeleOp** list › **Odometry Check** › INIT. **Don't touch the robot while pressing INIT.** Then press ▶. Gamepad 1: **CROSS** = start the test, **CIRCLE** = skip it. **Write down every result.**

1. Connected: passes by itself when the status is READY.

2. Stays put: press CROSS, then nobody touches the robot for 10 seconds.

FAIL → restart the OpMode, keeping the robot still during INIT.

3. Push forward: press CROSS, push the robot **forward about one foot** by hand without turning it, then press CROSS again.

"wrong way" → unscrew the **X pod**, turn it around 180° (still rolling forward), screw it back, repeat.

"swapped" → swap the two pod cables at the Pinpoint, repeat.

4. Push left: the same, pushing **left**. Same fixes, for the **Y pod**.

5. Spin: clear one tile all around the robot. Press CROSS: it turns twice to the left **by itself**.

Expected today: FAIL with "A or B off by about … mm", because A and B are not in the code yet. **Write the number down**; it's not a problem.

"Pinpoint and IMU disagree" → check the Pinpoint board is **flat, sticker side up**.

Step 4 · Measure A and B for Asif

1. **Find the robot's centre** and mark it with tape: halfway between the left and right wheels, and halfway between the front and back wheels.
2. **A = the X pod** (the one near the front wheel), in mm: how far **sideways** the middle of its wheel is from the centre. It sits near a side channel, so expect roughly half the robot's width. Standing **behind** the robot, looking forward: X pod on the **left** → **+A**; on the **right** → **−A**.
3. **B = the Y pod** (middle of a side channel), in mm: how far **forward** the middle of its wheel is from the centre. Expect a **small** number. **+B** if it's in front of the centre, **−B** if behind.
4. Write down: **A = ____ mm, B = ____ mm**. Also note if you turned a pod around or swapped cables in Step 3.

Step 5 · The robot drives by itself: Odometry Path Test

Driver Station › **Autonomous** list › **Odometry Path Test** › INIT. Don't touch the robot during INIT.

1. **Mark the start**: tape along the robot's **front edge and one side**.
2. **Path** (gamepad 1, D-pad ↑/↓). **Today only:**
   - **Square**: forward, strafe left, back, strafe right (4 ft).
   - **Diamond**: diagonals (4 ft ahead, 2 ft each side).

   Not yet "Square with turns" or "Field loop": they turn, so they need A and B.
3. **Laps** (D-pad ←/→): start with **1**.
4. **Space**: the screen shows how much room ahead, left and right the path needs.
5. **Let go of the gamepad, then press ▶.** The robot drives the path and comes back.
6. **Measure** how far it ended from the tape marks, in inches. **Goal: ½ inch or less.**
7. Order: Square 1 lap → Square 3 laps → Diamond 1 lap → Diamond 3 laps. Write down each result.

If the screen says	Do this
RUNAWAY or OUT_OF_BOUNDS	Redo Step 3, tests 3 and 4: a pod is probably reversed
TIMED_OUT	Check nothing is blocking the robot or rubbing on a wheel
NOT STARTED: not READY	Restart, keeping the robot still during INIT

Results for Asif

Check	Result
Pinpoint configured as **`pinpoint`** on I²C bus 1?	`yes / no`
Odometry Check 1 · Connected	`PASS / FAIL`
Odometry Check 2 · Stays put	`PASS / FAIL`
Odometry Check 3 · Push forward (what the screen said)	
Odometry Check 4 · Push left (what the screen said)	
Odometry Check 5 · Spin: “off by about … mm”	
A = ____ mm (X pod) · B = ____ mm (Y pod)	
Pod turned around? Cables swapped?	
Square · 1 lap: distance from tape	
Square · 3 laps: distance from tape	
Diamond · 1 lap: distance from tape	
Diamond · 3 laps: distance from tape	

Next: Asif puts A and B into the code (Constants.Odometry) and builds it. Then the spin test and the paths with turns, and finally the whole robot switches to the Pinpoint (USE_PINPOINT = true). Full install guide: docs/pinpoint-install.md.